VD detector protection

System roles, BDE hardware and the VD extension

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DEEP UNDERGROUND NEUTRINO EXPERIMENT

Fermilab

DPS arbitrates protection; SC operates, monitors and logs

DPS	**Detector Protection System.** Uses facility and detector conditions to allow or block actions; issues permits, inhibits and trips.
SC	**Slow controls, using Ignition.** Operates and monitors the detector; owns the data and event logs, including DPS decisions.
BMS	**Building Management System.** Controls building services. FFDG BMS and SDSTA BMS are separate systems.
Local interlocks	**Equipment-level protection.** Can trip without waiting for SC or the central DPS.

An operating request cannot override a protection trip.

DPS receives inputs from four system domains

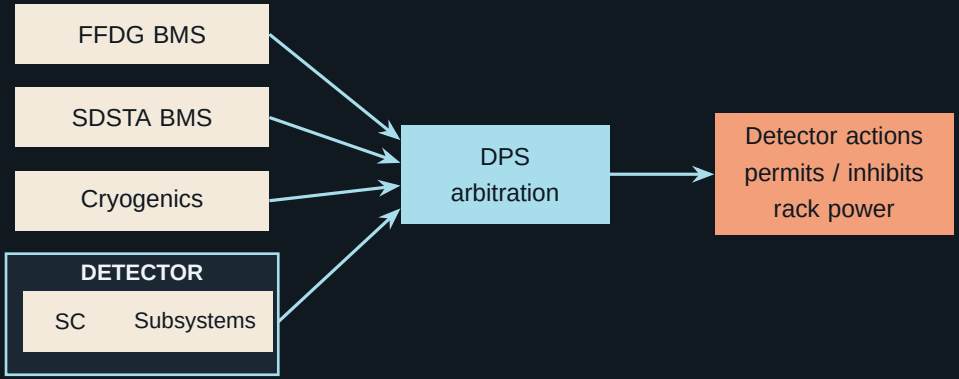


Diagram. Example: low liquid level withdraws the HV permit. SC logs the condition, decision and equipment state.

This is the logical system view. Facility and cryogenic systems retain control of their services. SDSTA retains personnel-safety responsibility.

Inside the detector: SC and the subsystems feed DPS

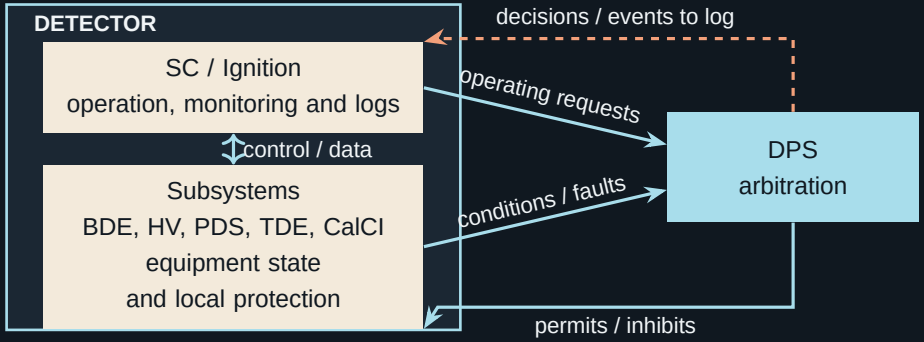


Diagram. SC owns the logs. DPS arbitrates protection across systems; the equipment retains its local interlocks.

HV = high voltage; PDS = photon detection system; TDE = top drift electronics; CalCI = calibration and cryogenic instrumentation. BDE is expanded next.

BDE already defines a distributed protection system

BDE = bottom drift electronics. I/O = inputs and outputs.

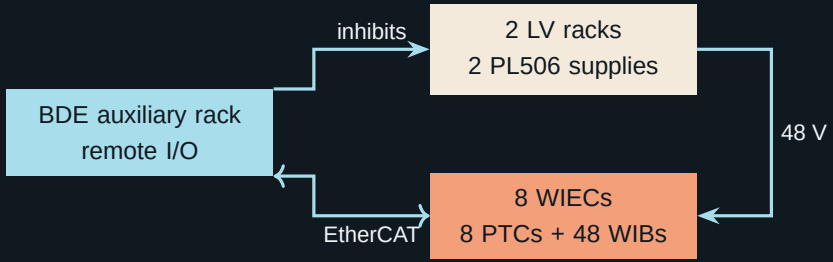


Diagram. Ten groups: 20 PL506 supplies, 80 WIECs and 480 WIBs.

The auxiliary I/O also connects fans, heaters and temperature sensors.

WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. June 2026 layout; fan/heater wiring omitted here.

BDE drawings, parts and software are already available

TPC protection panel designs

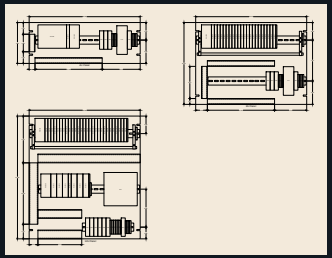
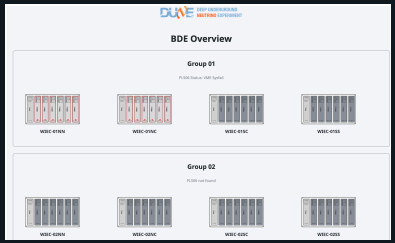


Diagram. Central PLC, bias I/O and LV/fan/heater panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

TPC reference: EDMS 2401090 v4, sheet 2, with parts and power workbooks. PLC = programmable logic controller.

Proposal: extend the BDE PLC design to the rest of VD

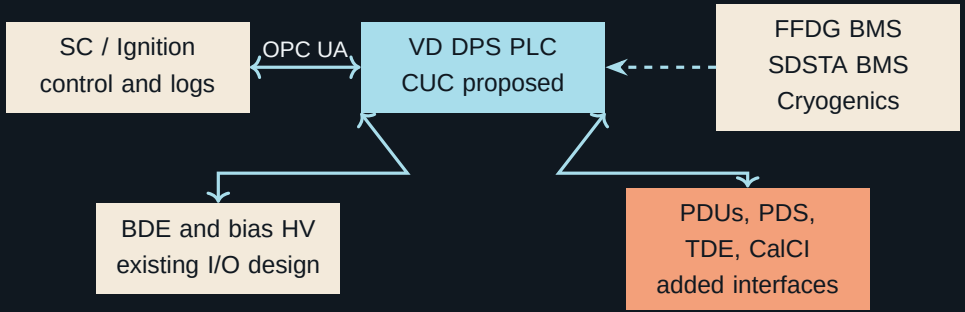


Diagram. Place the central PLC beside the other PLC racks in the CUC. Keep distributed I/O near the equipment.

PLC = programmable logic controller; CUC = Central Utility Cavern. Dashed link: select the hardware/software path for each facility input. HD can later use an extension or a separate PLC.

Ignition controls the PLC; the PLC controls the PDU

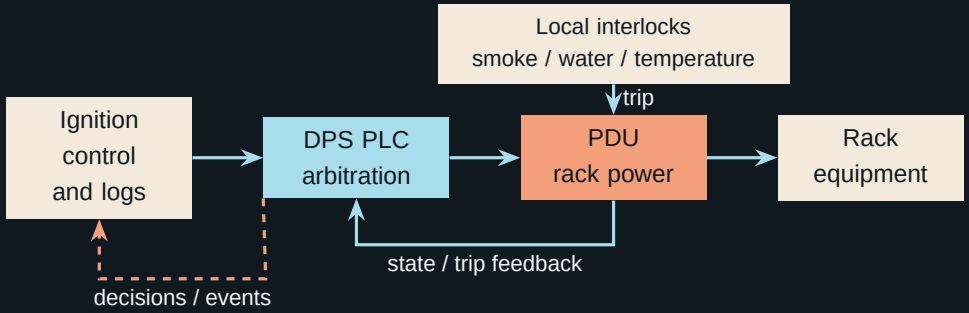


Diagram. The PLC arbitrates Ignition's commands. Local interlocks can trip independently; SC logs the outcome.

PDU = power distribution unit. An Ignition command cannot override a trip. Reset and re-enable are separate actions.

The VD extension adds subsystem and facility interfaces

VD scope	Added DPS interface
PDS: 34 LV feeds; 80 PoF boxes	Supply control, fault feedback and local PoF readout/control
TDE: 20 racks	Crate-status exchange and external inhibit interface
Drift HV / CalCI	Level, temperature and optical-source status; inhibits and operation permits
Facility interfaces	FFDG BMS, SDSTA BMS and cryogenic conditions for DPS arbitration

The 80 PoF boxes have 1,280 monitoring values. Local readout handles the measurements; protection permits and feedback determine the DPS I/O.

LV = low voltage; PoF = power over fibre. Parts, panel allocations and power estimates are in the backup slides.

Test arbitration, trips and recovery on the bench

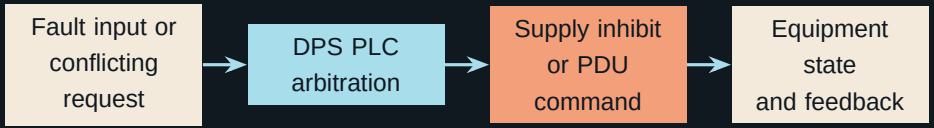


Diagram. Use the BDE vertical-slice work and G271 equipment for representative VD paths.

Measure detection through completed shutdown. Test local trips, SC and fieldbus loss, reset and re-enable. Verify that SC logs the decisions and equipment states.

No critical sub-ms function has been identified so far. Simulation prepares the fault and load cases; hardware tests measure the complete response.

Take the VD hardware to review in June 2027

Work before review	Result for procurement
Adapt the BDE design	VD panels, added I/O, CUC fieldbus routes and power allocation
Test the interfaces	Trip-time, failure and recovery results; SC/PDU integration
Prepare purchases	Drawings, parts and cable lists, costs, lead times and spares

After readiness review: order the hardware and install it in the site sequence.

Possible facility-interface need: summer 2027. Full VD response-matrix review: early 2028. The early matrix work supplies requirements that affect hardware purchases.

I will lead the VD hardware work with a control engineer

Proposed contribution	Work
Manuel + control engineer	VD design, SURF interfaces and procurement package
Terri and Trevor	BDE panels, PLC/I/O implementation and bench work
Geoff + Manuel	SC commands, PDU protection and recovery
Vivian + consortia / DPS	Equipment requirements and ICD updates

Support needed: control-engineering time and access to the existing test stands.

SURF work: inspection maps are ready; access is being coordinated with Rachel, with Dale Curran identified as an SDSTA integration contact.

Backup: VD rack space already allocated to DPS

Rack type	Count	DPS equipment / connections
BDE auxiliary	10	12U per rack; eight WIEC connections per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U per rack for the DPS interface to ISEG supplies

The auxiliary racks contain the WIEC I/O; the bias racks contain the ISEG inhibit outputs.

May 2025 VD rack workbook; June 2026 BDE layout. U is rack height. These are design allocations; the bias-HV sheet is a first estimate.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. Quantities describe the first-pass panel layout.

VD power estimates for the BDE racks

Equipment	Per rack	VD basis
PL506 input	2.25 kW	20 racks: about 45.1 kW for the WIEC supplies
Auxiliary DPS input	About 1.11 kW	10 racks: about 11.1 kW; eight WIECs per rack
Bias DPS supply allowance	24 V, 40 A	960 W DC capacity per rack in the workbook

The auxiliary estimate scales the recorded 831 W from six to eight WIECs: $831 \times 8/6 = 1{,}108$ W.

Planning estimates from the May 2025 workbook. Linear scaling follows its VD note; it is not a measured load. Supply capacity is separate from AC demand.

Reference: local PDU trips and recovery

PX3 setup guide	Local behaviour documented
Smoke, water, temperature	Local event rules and scripts can switch off the protected outlets
Network-switch outlet	Kept powered during the guide's fallback shutdown
After a local trip	Power is restored deliberately; no automatic restoration

The VD control chain is Ignition → DPS PLC → PDU. Local trip status returns to the PLC and to SC for logging.

DUNE-doc-32655 v8, pp. 3–6 and 14–16, supplies the reference local functions. The control chain follows the VD architecture.

The VD inventory covers the other subsystem connections

Subsystem	Working inventory and interfaces
PDS readout / calibration	34 LV feeds in the current presentation scope; supply status and control interfaces
Power over fibre (PoF)	80 laser boxes; 640 temperature and 640 voltage readings; compact PLC interface proposed
TDE	20 mezzanine racks; power supplies and readout-crate management
Calibration / cryogenic instrumentation	Level and temperature readout, purity monitors, ionization laser and neutron source

VD allocation. Proposed routing: monitoring to Ignition through native drivers and OPC UA; protection permits and fault status exchanged with the local equipment.

The BMS drawing identifies 20 control-panel symbols

Drawing area	Interfaces on the inspection maps
CUC, level 4850	Eight panel references: air handling, chilled water, power and associated equipment
Level 4910	North and south sump interfaces, SP-1 and SP-2
4850-08-ES group	Spare panel for future equipment; mapped to the access-drift inspection area
Other 4850 areas	Chillers, pumps, ventilation and generator interfaces

Inspection planned with Rachel; Dale Curran is the SDSTA integration contact identified for this work.

U-FD-M-901 rev12, 14 February 2025; marked 4850/4910 maps and survey register. Symbols are drawing references; field observations are pending.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

May 2025 allocation: rack positions, distinct from PLC panels and remote-I/O stations. Includes the listed reserved and spare positions.